

Development of Endocrine system

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Pituitary gland (Hypophysis cerebri)

developed from the ectodermal tissue.

RATHKE'S POUCH- epithelial in origin from the roof of mouth (oral ectoderm)

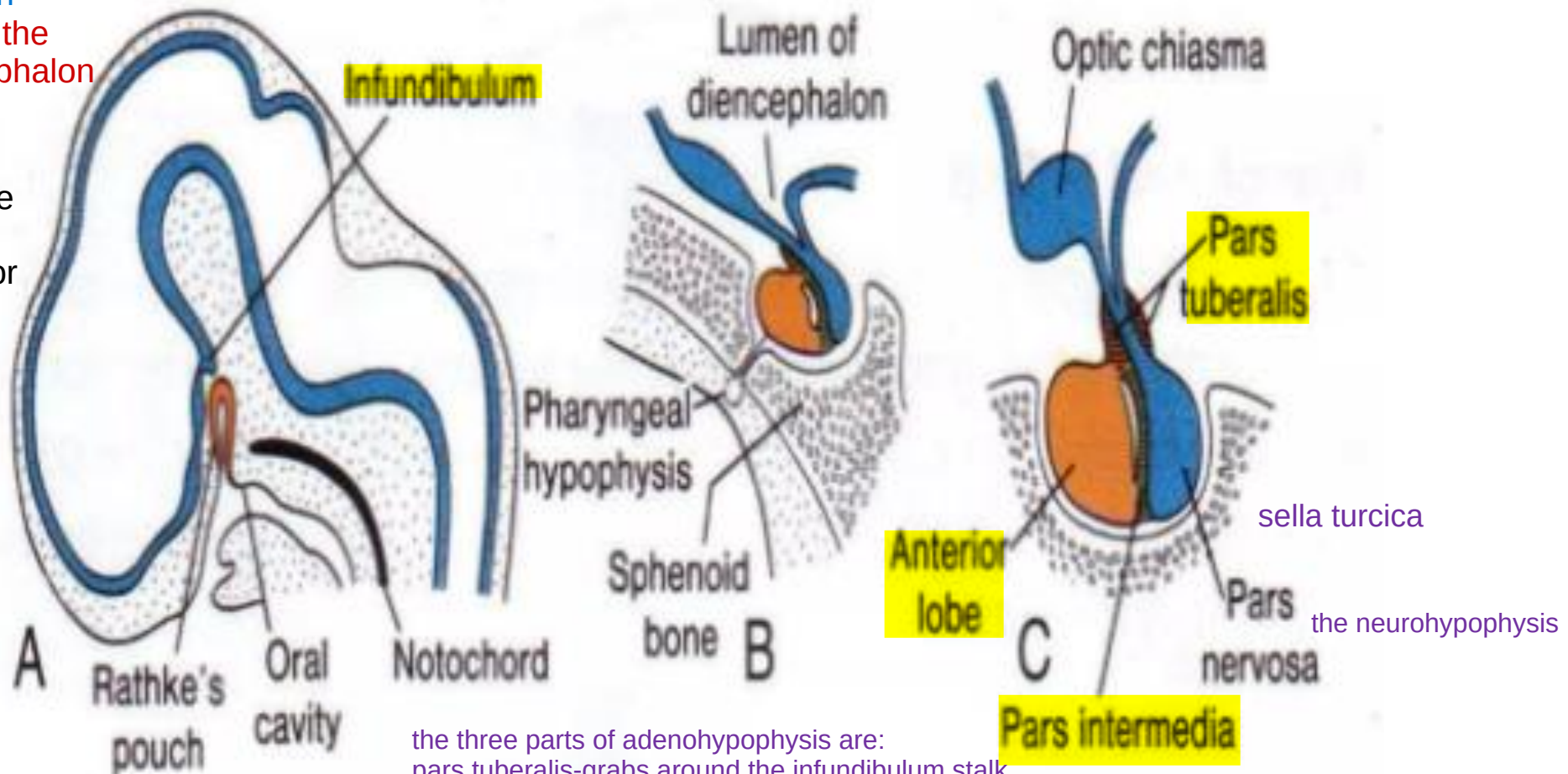
INFUNDIBULUM-neural ectoderm from the ventral diencephalon

it is attached to the undersurface of the brain by infundibulum(stalk)
hypophysis is a Greek word meaning attachment underneath

The third ventricle= the cavity of the diencephalon

ectoderm gives rise to either surface ectoderm, neural tube or neural crest cells

neural ectoderm= neural tube + neural crest cells



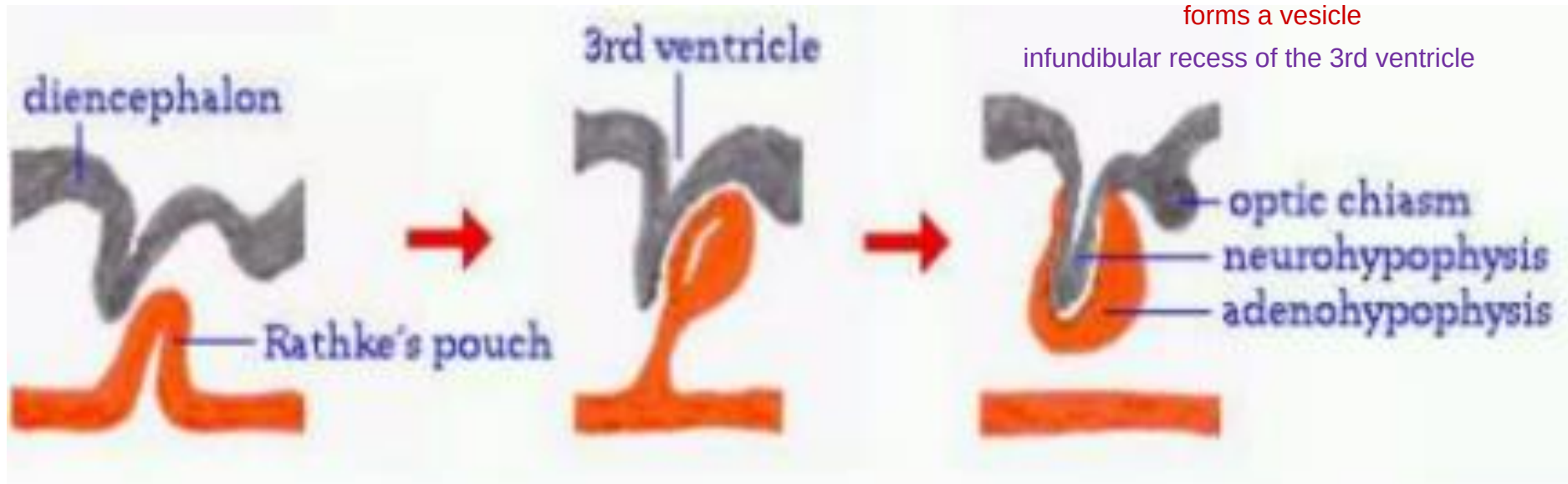
the three parts of adenohypophysis are:
pars tuberalis-grabs around the infundibulum stalk
pars disatlis-the main part of anterior lobe
pars intermedia- the intermediate lobe of PG

Adenohypophysis

anterior lobe of the pituitary gland

- Develops from an evagination of ectoderm lining, the roof of primitive oral cavity (stomodeum) → Rathke's pouch....4th wk.

Rathke's pouch detaches itself from the roof of the mouth and forms a vesicle



pars anterior = pars tuberalis + pars distalis

The cells of anterior wall of Rathke's pouch undergo extensive proliferation to form the anterior lobe (pars anterior)

The posterior wall proliferates more slowly to form the intermediate lobe (pars intermedia)

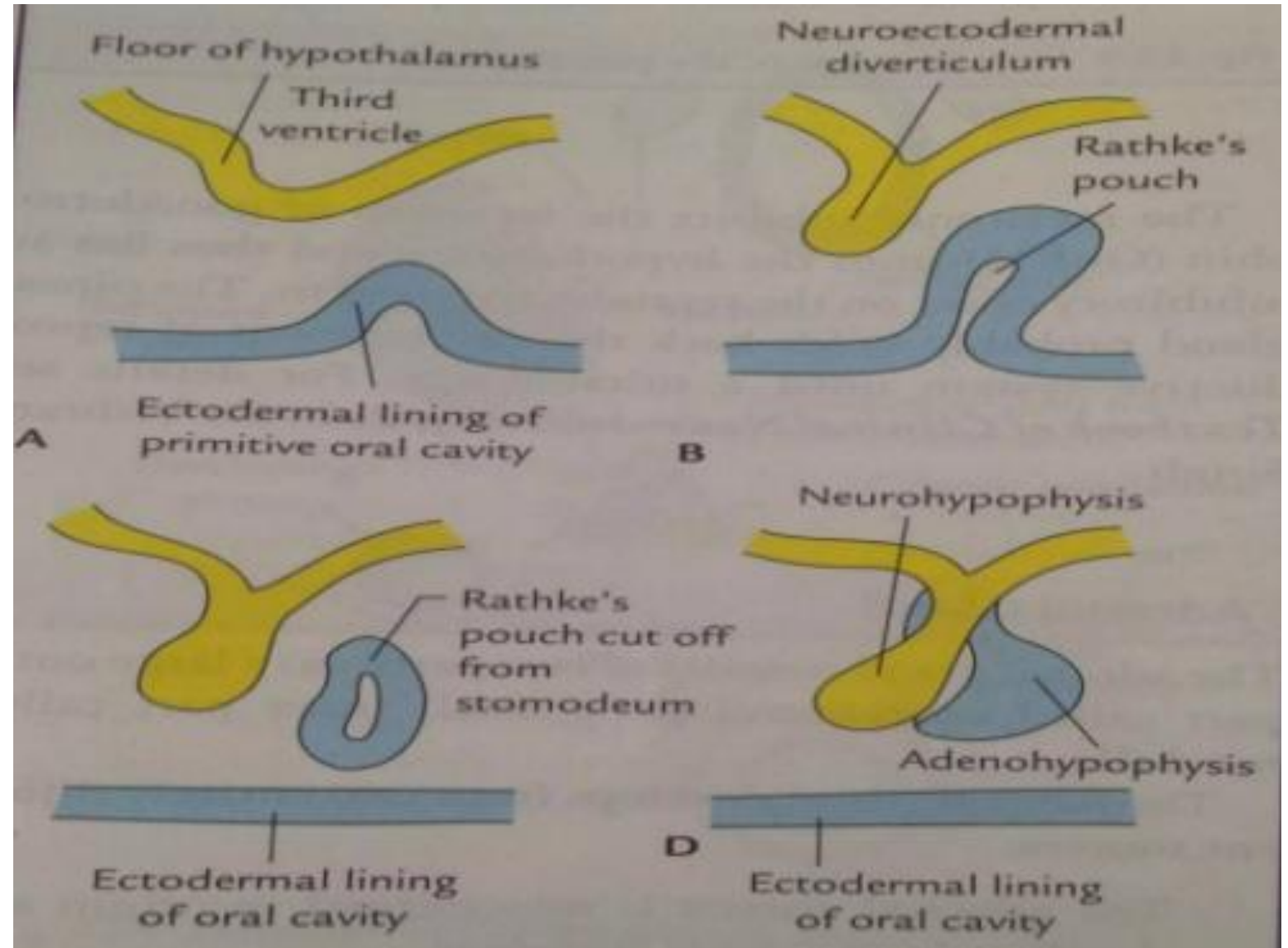
The space between pars intermedia and pars anterior is called pituitary cleft

Neurohypophysis

posterior pituitary= pars nervosa + infundibular stalk

posterior lobe of pituitary gland
the infundibulum gives the posterior lobe

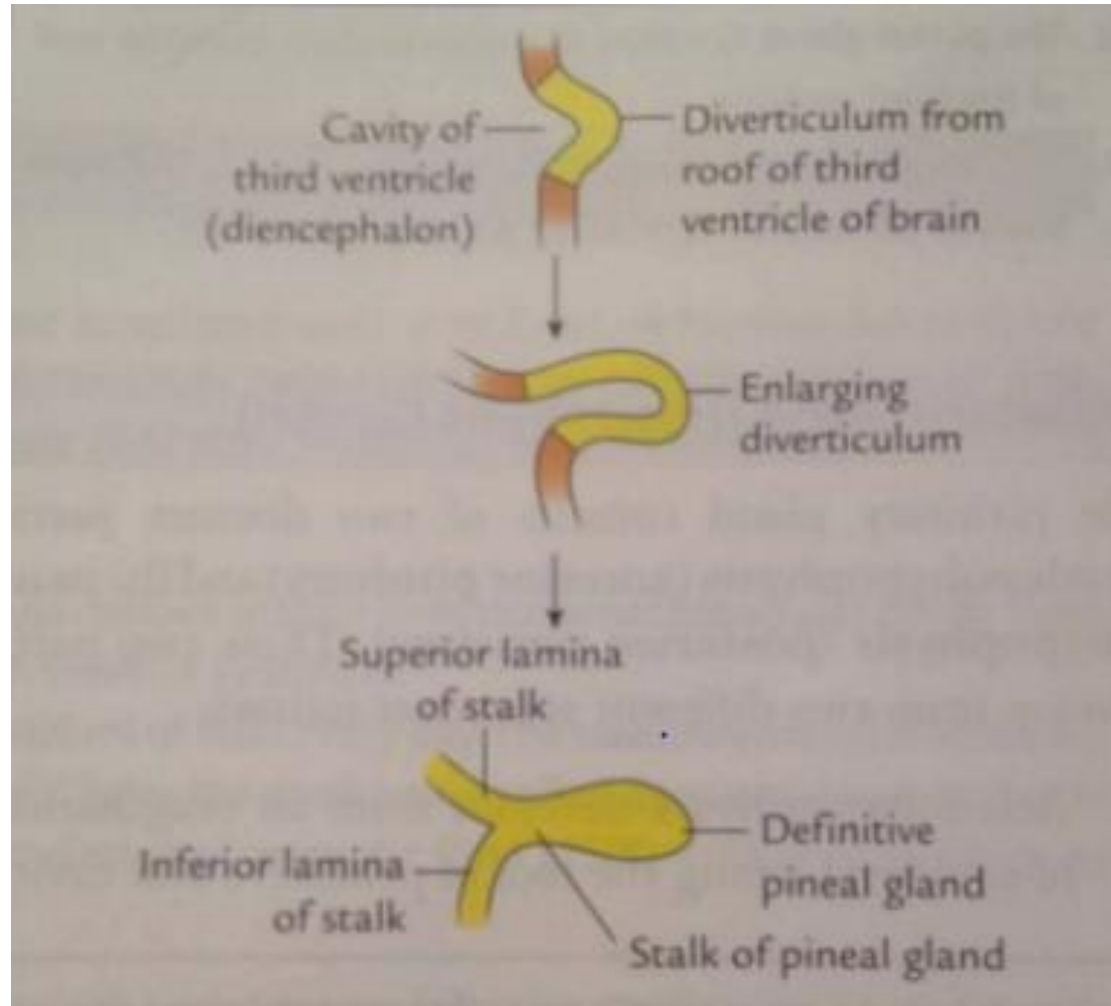
- Develops from an evagination of neuroectoderm of diencephalon (floor of 3rd ventricle)
- Fused with from the Rathke's pouch → Hypophysis cerebri



Pineal gland (Epiphysis cerebri)

- Develops from the **roof of the diencephalon**
purely ectodermal in origin (from the neural ectoderm)

Ectopic posterior pituitary is a condition in which the back end (posterior) of the pituitary gland is in an abnormal position.



The pharyngeal hypophysis is a small piece of extra sellar pituitary tissue derived from the stalk of Rathke's pouch and is situated in the roof of the nasopharynx

A craniopharyngioma is a rare type of brain tumor derived from pituitary gland embryonic tissue that occurs most commonly in children, but also affects adults

Thyroid gland

the secretory cells(follicular cells T3 and T4 secreting cells) are endodermal in origin but the parafollicular cells(few but relatively larger cells that secrete calcitonin) rise from the neural crest cells that infiltrate into the 4th pharyngeal pouch

the current site of foramen cecum

- In the midline of the floor of the pharynx, the endodermal lining of the foregut forms the **thyroid**

diverticulum during the 4th week of embryonic development

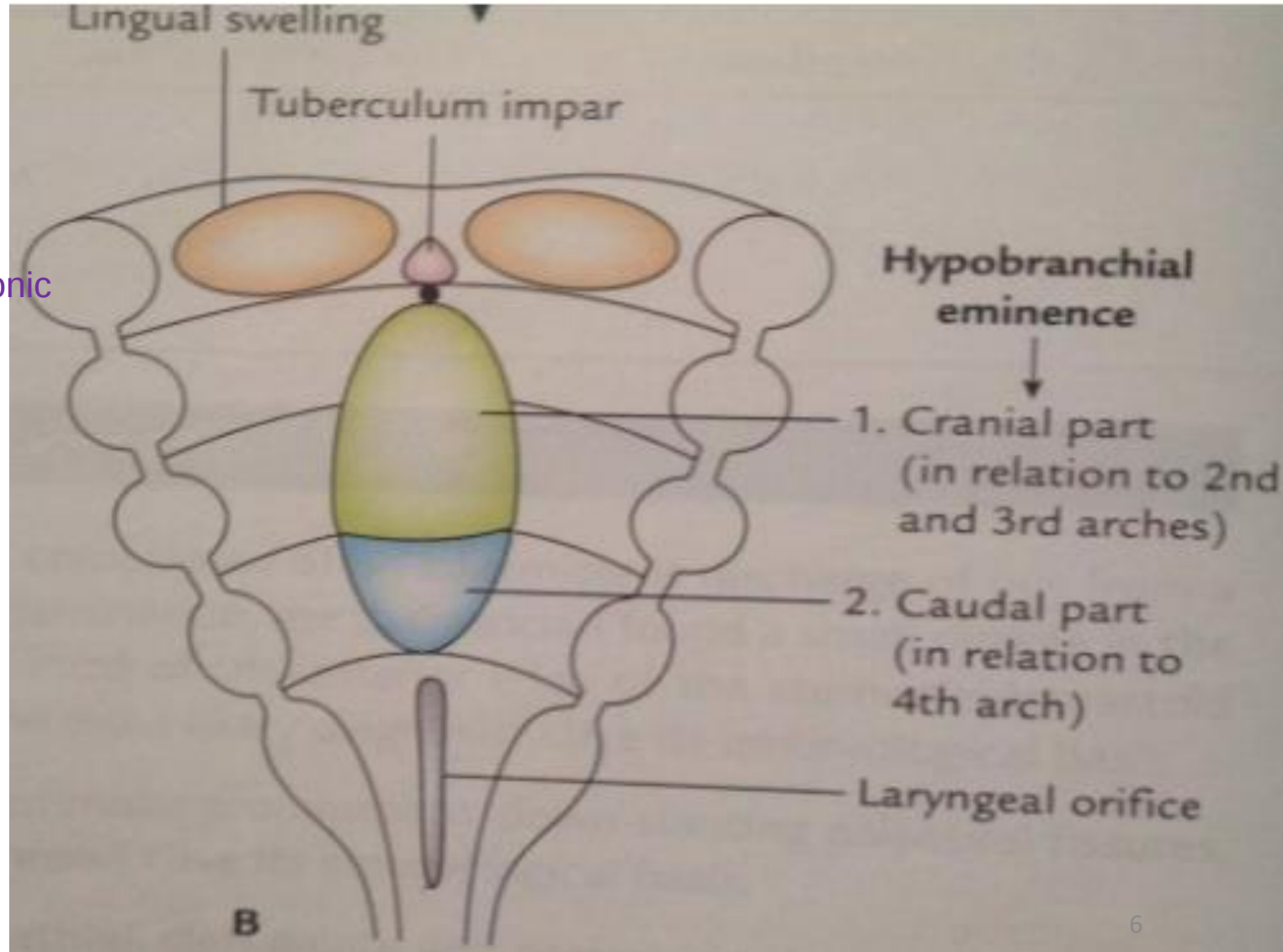
- Migrates caudally, passing ventral to the hyoid bone and laryngeal

cartilages during the 7th week of embryonic development

starts functioning approximately at the end of third month

- Remains connected to the tongue by the **thyroglossal duct**, which later is obliterated

8/30/2023 there is no connection after 7th week



Cont....

the stromal cells/the supporting cells are mesodermal in origin while the follicular cells/the secretory cells are endodermal in origin

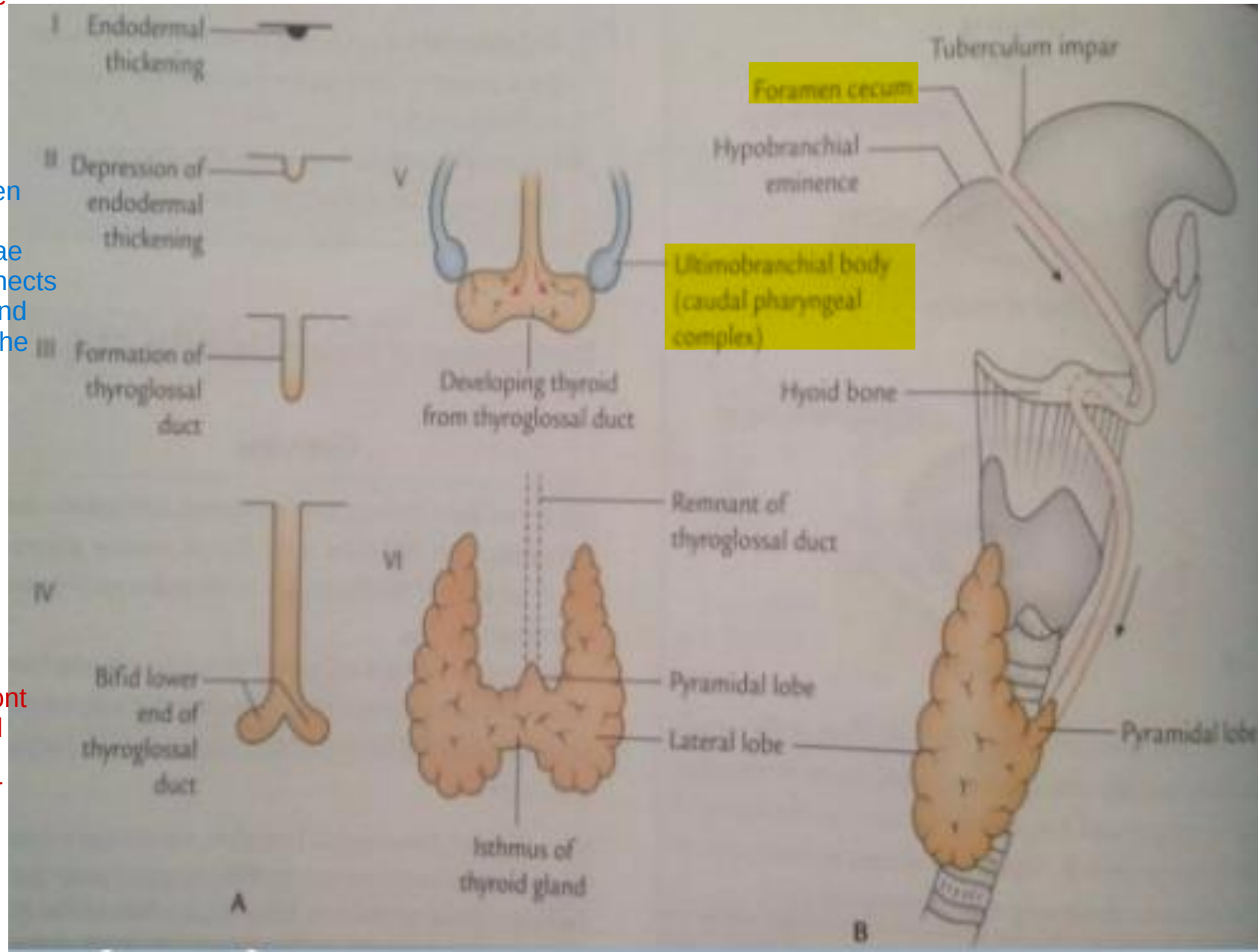
Thyroglossal duct originates at the junction of the anterior two-thirds and the posterior third of the tongue at its base(foramen cecum).it connects the thyroid diverticulum to the tongue

the remnant of thyroglossal duct proximally forms foramen cecum and distally forms the levator glandulae thyroideae fibromuscular band that connects pyramidal lobe of thyroid gland to the hyoid bone where as the middle part obliterates

Lingual thyroid is a mass of ectopic thyroid tissue, located at the base of the tongue, caused by a developmental anomaly of the descent of thyroid gland

The thyroid gland is normally located at the front of the neck. A retrosternal thyroid refers to the abnormal location of all or part of the thyroid gland below the breastbone (sternum)

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A thyroglossal duct cyst is a mass or lump in the front part of neck that is filled with fluid. A thyroglossal cyst is a fibrous cyst that forms from a persistent thyroglossal duct.

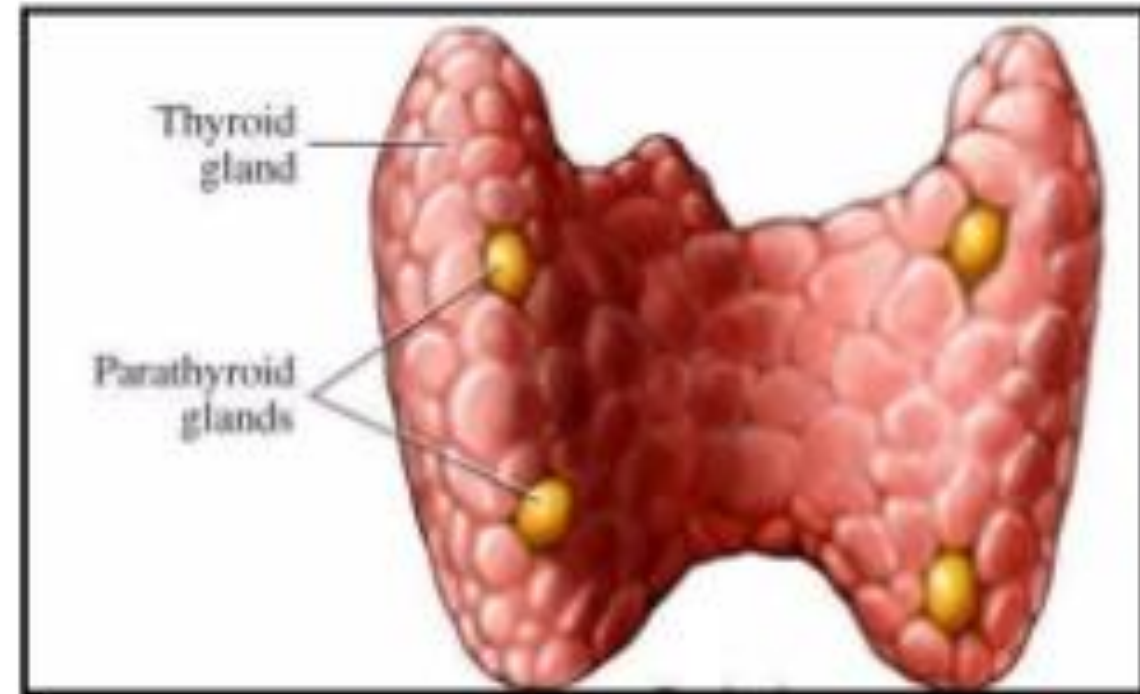
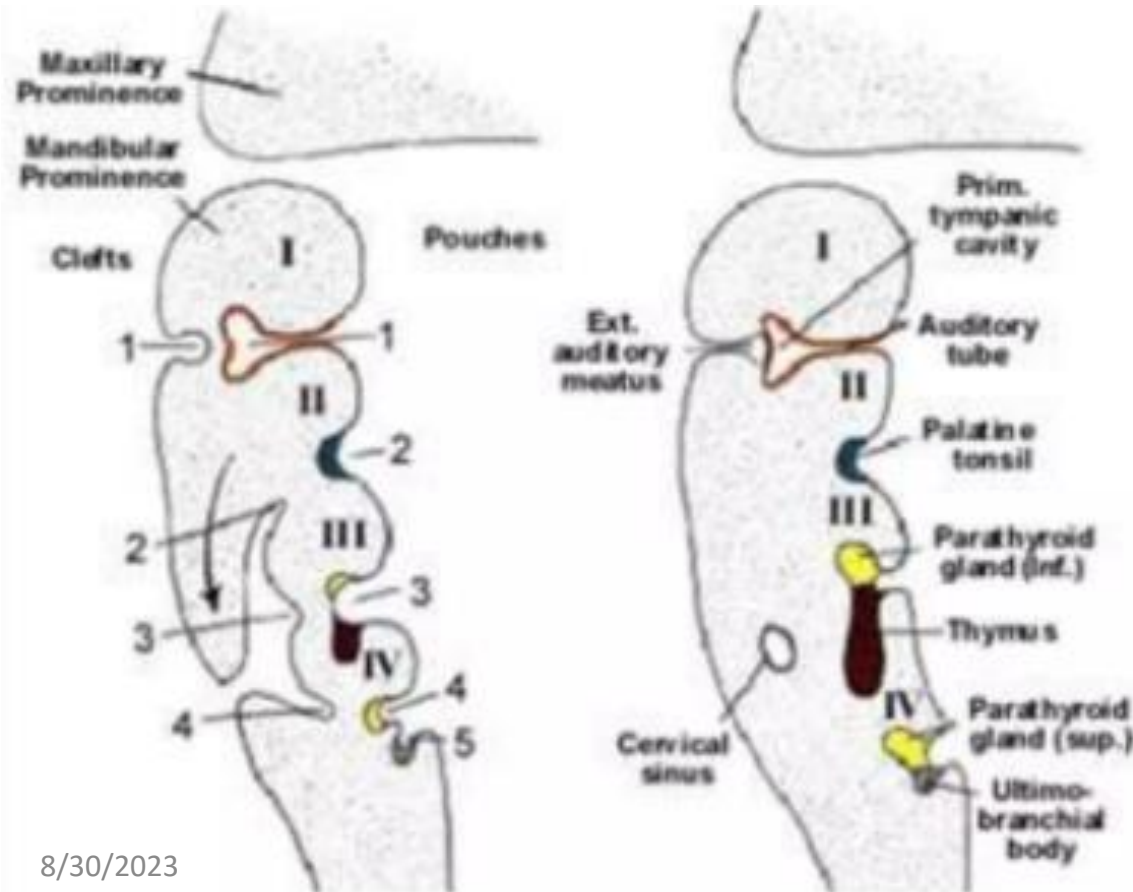
Parathyroid glands

endodermal in origin

the inferior parathyroid gland develops from the 3rd dorsal pharyngeal pouch
the superior parathyroid gland develops from the 4th pharyngeal pouch

- Develop from 3rd and 4th pharyngeal pouches

there are two types of cells chief cells(that secrete parathyroid hormone)
and oxyphil cells whose function is unknown.



Suprarenal gland

resembles the postganglionic sympathetic neurons

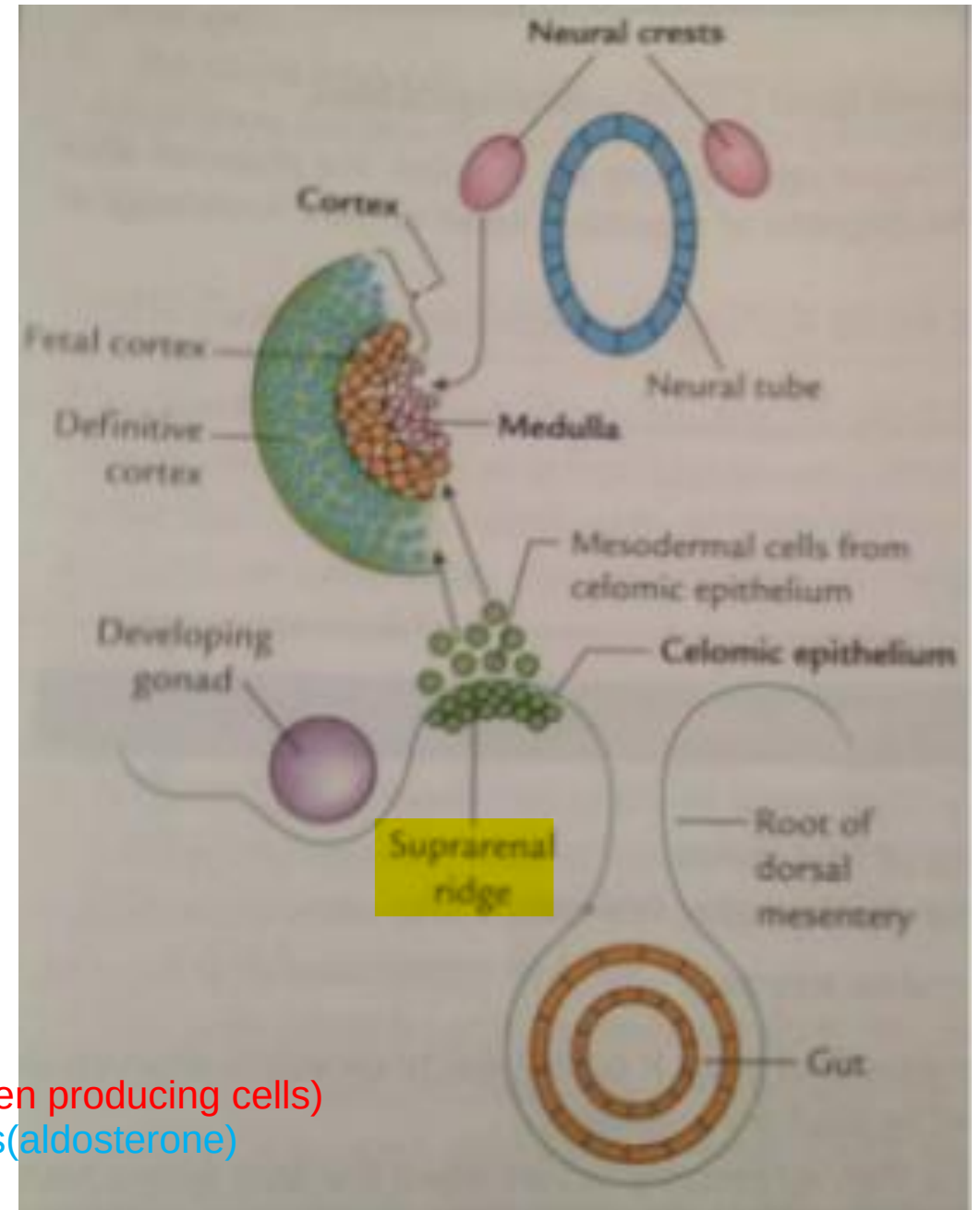
- **Cortex** → **coelomic epithelium**
mesodermal origin
accounts for 80% of the gland
- Fetal cortex → larger acidophilic cells
- Definitive cortex → smaller cells

- **Medulla** → **neural crest cells**
ectodermal origin

made up of chromaffin cells, the postganglionic neurons that produce epinephrine norepinephrin catecholamin

the cortex has three parts zona granulosa zona reticularis(androgen producing cells)
zona fasciculata all producing steroid hormone, mineralocorticoids(aldosterone)
glucocorticoids.

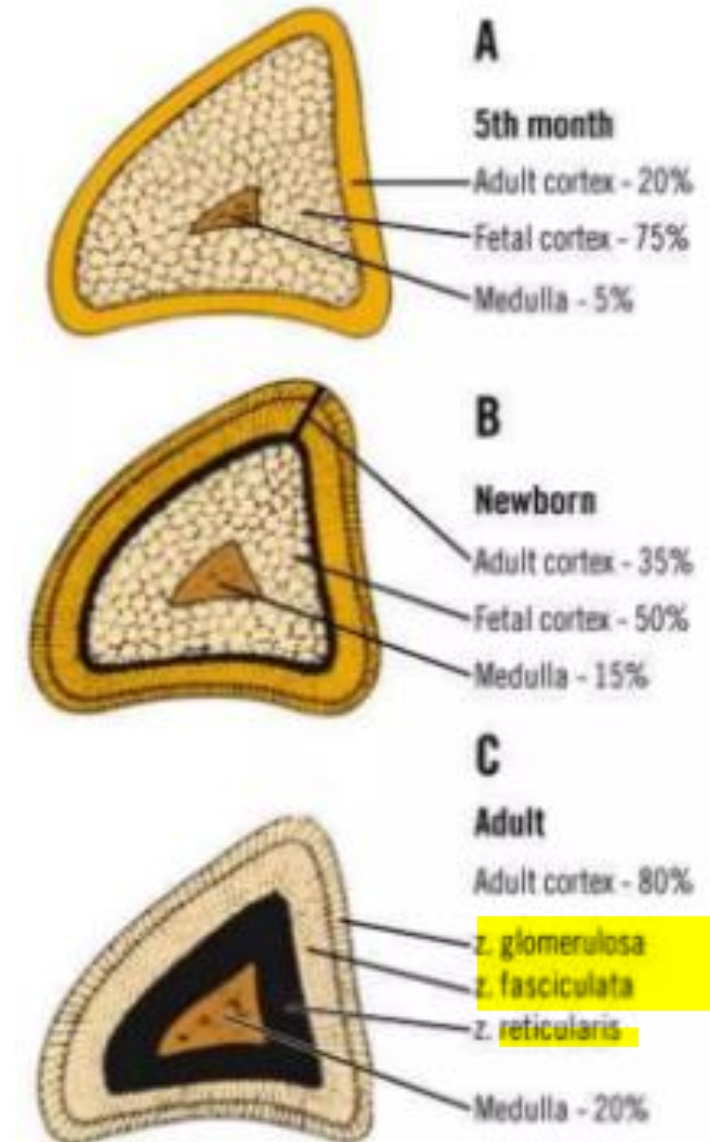
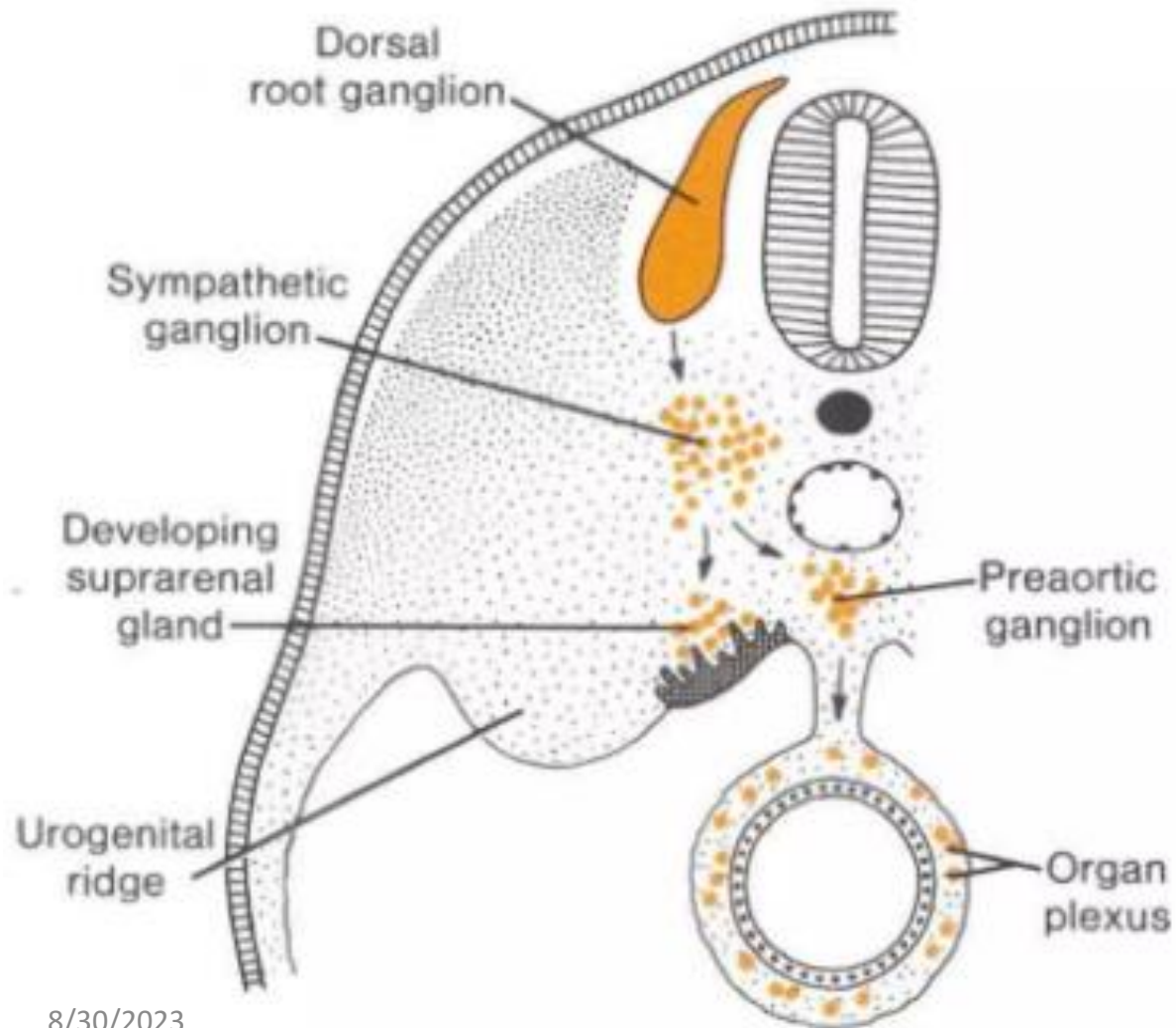
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Cont....

hypothalamus is ectodermal in origin(neural ectoderm)

pancreatic cells(both endocrine and exocrine cells)
has endodermal origin.



Congenital anomalies: Reading assignment

Cushing's syndrome is a disorder that occurs when your body makes too much of the hormone cortisol over a long period of time.

Adrenogenital syndrome is also known as congenital adrenal hyperplasia, which results from disorders of steroid hormone production in the adrenal glands leading to a deficiency of cortisol.

Pheochromocytoma is a type of neuroendocrine tumor that grows from cells called chromaffin cells

Chromaffin cells are neuroendocrine cells found predominantly in the medulla of the adrenal gland

- Pituitary agenesis/hypogenesis impaired formation of pituitary gland
- Ectopic posterior gland when pituitary gland is found in the base of the brain other than sella turcica (the original normal place) AKA pituitary dwarfism
- Pharyngeal hypophysis ectopic or accessory pituitary gland found at the roof of the pharynx when the cranio-pharyngeal canal fails to close
- Craniopharyngioma tumor of pituitary gland
- Lingual thyroid
- Retrosternal thyroid excess descent of thyroid to the superior mediastinum
- Thyroglossal cyst remnant of thyroglossal duct forms cyst along the course anterior to hyoid bone or anterior to larynx
- Parathyroid agenesis/hypogenesis non formation or impaired formation
- Conn's syndrome & Addison's disease
- Cushing syndrome
- Adrenogenital syndrome
- pheochromocytoma

Conn's syndrome is a hormonal condition in which one or both adrenal glands produce more of the hormone aldosterone than normal

Addison's disease, also called adrenal insufficiency, is an uncommon illness that occurs when the body doesn't make enough of certain hormones.

*Thank
you*

